



# SPOT-MP.AI

Intelligent Microplastic Awareness





Agentic AI Hackathon 2025  
Organized by Ulster University, UK



**SPOT-MP.AI**  
Intelligent Microplastic Awareness

|                |   |                                                                                                                                                                                                       |
|----------------|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Team Number    | : | T-1998                                                                                                                                                                                                |
| Team Name      | : | <b>SPOT-MP.AI</b>   Intelligent Microplastic Awareness<br>( <b>S</b> mart <b>P</b> latform for <b>O</b> ptical <b>T</b> racking of <b>M</b> icro <b>P</b> lastics using <b>AI</b> )                   |
| Problem Theme  | : | <b>Healthcare &amp; Life Sciences</b><br>Turning invisible contamination into visible knowledge for public awareness and health protection.                                                           |
| Institute Name | : | <ul style="list-style-type: none"><li>King Fahd University of Petroleum and Minerals (KFUPM)</li><li>Denmark Technical University (DTU)</li></ul>                                                     |
| Team Members   | : | <b>Nur Allif Fathurrahman</b> , Esraa Kotob Abas, Listya Eka Anggraini, Sharhan Hasabi Haputra, Izzul Qudsi, Anisa Helena Isma Putri                                                                  |
| Contact        | : | <a href="mailto:g202308910@kfupm.edu.sa">g202308910@kfupm.edu.sa</a> or <a href="mailto:allif.fathur@gmail.com">allif.fathur@gmail.com</a><br>+966 55-857-7165 (Saudi); +62 877-7003-0767 (Indonesia) |

Date: 8 Oct 2025

# Microplastics: The Hidden Threat in Our Food, Water, and Air

Invisible Microplastic Exposure

**>90%**

of bottled water, seafood, packaged snacks, beverages, fruits, and vegetables, and household dust inhalation contain microplastics



Lack of Accessible Detection & Awareness

**0** public tools

to visualize or monitor microplastic presence and detection still depends on slow, expensive lab analysis



Cost of Inaction

**5+ trillion**

plastic particles continue circulating in oceans and food chains

*Without awareness, contamination spreads unchecked, endangering future health and ecosystems.*

# Problem Statements

Consumers lack awareness, fast, and affordable tools to detect microplastics in their food and drinks.

**88%**  
*unaware*

## **Low consumer awareness**

Most people are unaware of how often microplastics contaminate food, drinks, and the environment, and their health impact

**60%**

*No tracing system*

## **Manufacturers are struggling in microplastics tracing**

Food and beverage producers lack effective ways to trace microplastic contamination across their supply chain

**\$150**

*Costing per sample*

## **No affordable microplastics detection tools**

Existing microplastic lab-based tests are costly and time-consuming for both consumer or industry use.

*If you can't see it, you can't solve it.*

# From Detection to Awareness

Transforming microplastic detection into real-time awareness, everyday action, and scalable environmental health impact.



**SPOT-MP.AI**  
Intelligent Microplastic Awareness

## Our Objectives



### AI-Powered Microplastic Detection

Develop an intelligent image-based system capable of identifying and classifying microplastics in food and water samples using fluorescence signatures and AI vision.



### Awareness and Visualization Platform

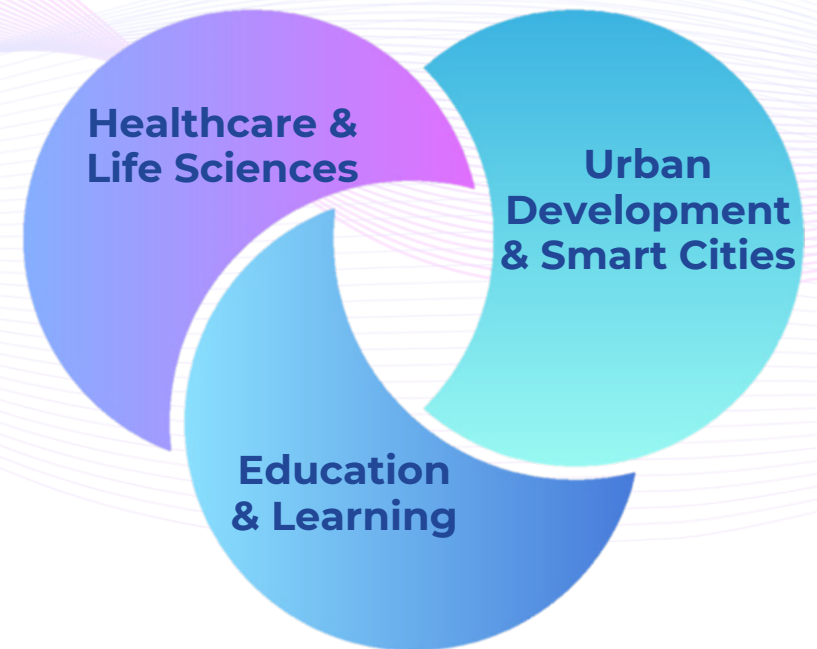
Build a user-friendly digital platform that visualizes detection results, tracks contamination data, and communicates real-time microplastic awareness to the public.



### Scalable and Affordable Monitoring Tool

Design a cost-effective, portable detection kit integrated with the SPOT-MP.AI platform to democratize microplastic testing for households, schools, and small labs.

## Strong Development Pillars





# SPOT-MP.AI

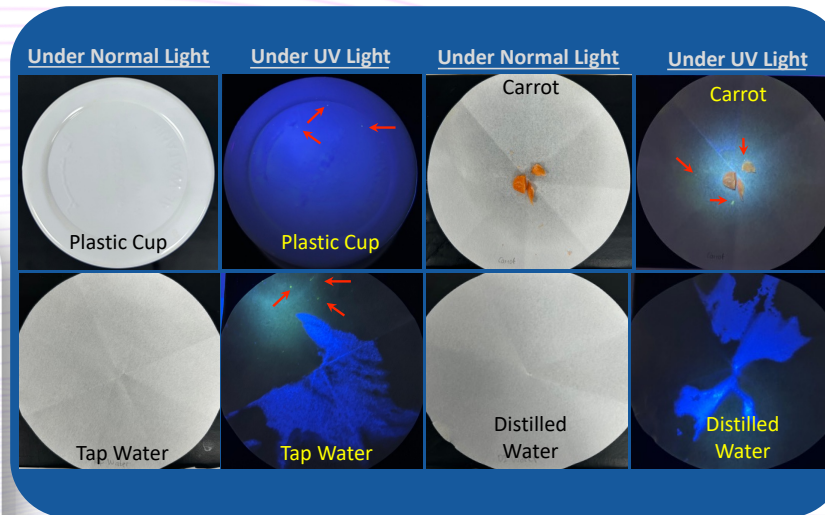
Intelligent Microplastic Awareness

Transforms microplastic detection into intelligent, real-time awareness using AI vision and fluorescence imaging.



## Non-Toxic and Bio-based UV Home-Test Kit

Simple, affordable, and easy-to-use tools for everyday microplastic detection



## Test Result from Sample

Bright-yellow fluorescence appears instantly as curcumin binds to hydrophobic microplastics and AgNPs amplify the signal under UV (365 nm)



## AI Vision Detection and Awareness Dashboard

Our trained model analyzes the image and identifies microplastic presence, size, level, and location

✓ Results in < 5 minutes (and only <\$20) instead of 24–48 hours lab time.

✓ Proprietary AI dataset trained on fluorescence microplastic images.

✓ Validated on 10+ fluorescence image sets with >85% model accuracy.

✓ MVP ready for pilot in schools, labs, and awareness programs.

# When Chemistry Meets Intelligence Awareness Becomes Action

SPOT-MP.AI — The World's First AI-Augmented Fluorescence Platform for Microplastic Awareness

**AI-Enhanced  
Fluorescence  
Detection**

**Real-Time  
Agentic  
Awareness**

**Accessible,  
Affordable,  
Autonomous**

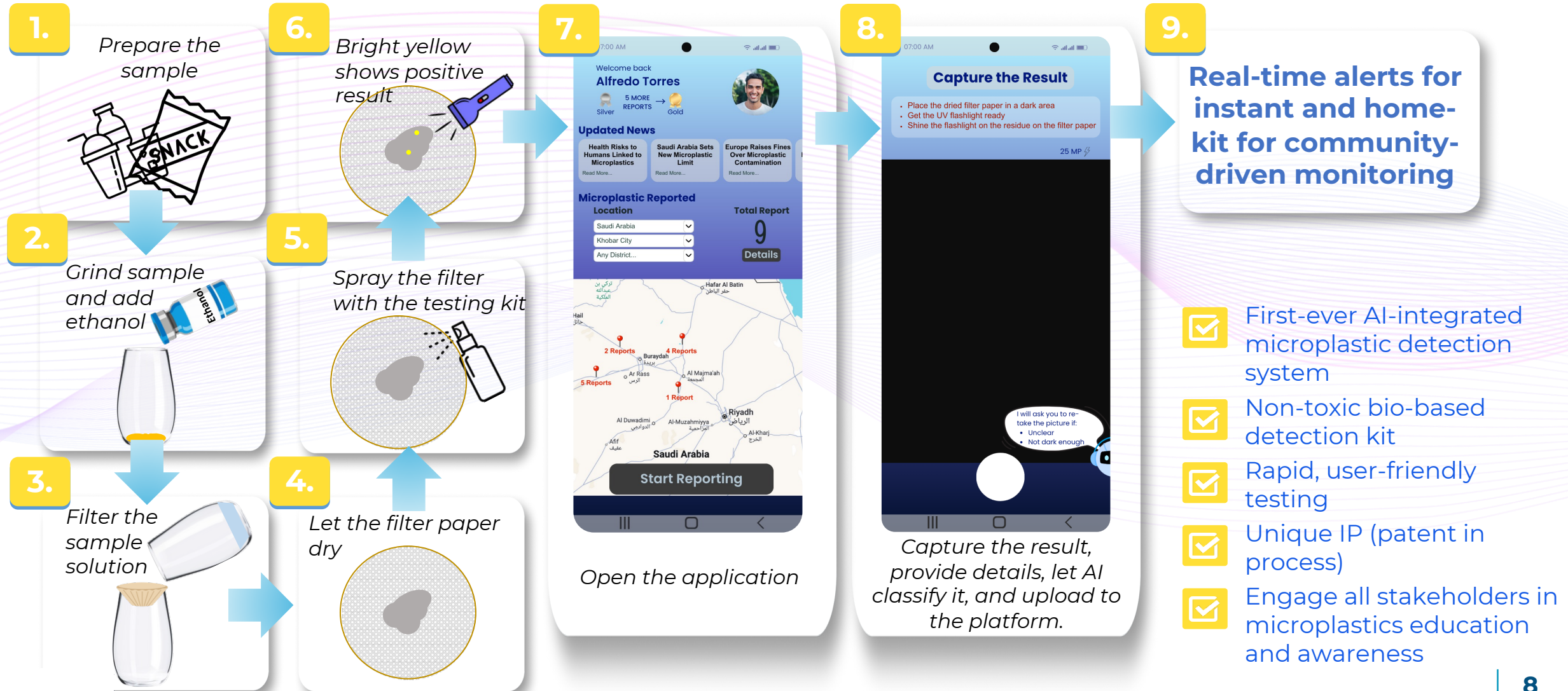
**Human + Machine  
Collaboration**

A single fluorescent glow under UV becomes a signal to the AI, proof that awareness can evolve intelligently, collectively, and globally.

*From microplastics to macro-impact, AI that learns from every drop, every detection, every human.*

# Intelligent Microplastic Detection Workflow

A seamless detection, classification, and education workflow with a clear show-stopper



# Tech Stack Used and Budget

Our tech stack fuses fluorescence chemistry, AI vision, and human-guided intelligence, built to scale within a \$15 prototype.

| Layer                        | Technology / Tools (2025 Verified)                                       |
|------------------------------|--------------------------------------------------------------------------|
| AI Vision & Detection        | YOLOv8 (Ultralytics), OpenCV 4.9, PyTorch 2.3, scikit-image              |
| Fluorescence Signal Analysis | Python NumPy + SciPy, ImageJ macro plugins, Matplotlib                   |
| Agentic AI Layer             | LangChain, GPT-4o API, Streamlit Agents                                  |
| Backend & Deployment         | Flask (3.0) + Firebase Auth + AWS EC2 Free Tier + FastAPI                |
| UI/UX & Visualization        | Plotly Dash, Streamlit, TailwindCSS, Figma                               |
| Hardware Interface           | Smartphone camera + UV 365 nm LED torch + light sensor API (Android/iOS) |
| Data Management              | Pandas + Labellmg + Google Drive API + GitHub                            |

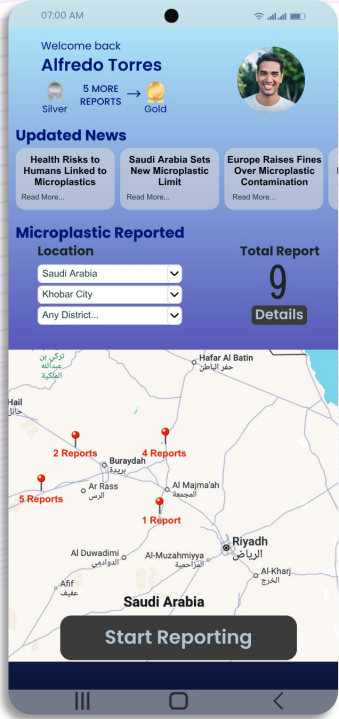
| Component                           | Unit Cost (USD)                                   |
|-------------------------------------|---------------------------------------------------|
| Spray formulation (100 mL)          | \$4.90                                            |
| Ethanol 99% (250 mL)                | \$1.30                                            |
| Plastic bottle + atomizer           | \$0.45                                            |
| UV flashlight 365 nm                | \$7.90                                            |
| Filter paper + consumables          | \$0.60                                            |
| Mobile sensor API & hosting         | \$0.00 Android/iOS + AWS Free Tier                |
| Web & AI integration                | \$0.00<br>GitHub + Streamlit Cloud + Hugging Face |
| Total Estimated Cost per Kit (Avg.) | ≈ <b>\$15.15 USD</b>                              |

# Deliverable / Expected Outcome

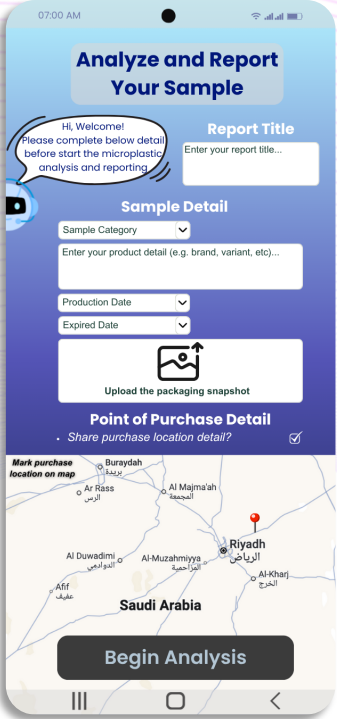
From Fluorescent Science to Agentic Intelligence

Delivering the world's first AI-powered platform for intelligent microplastic awareness.

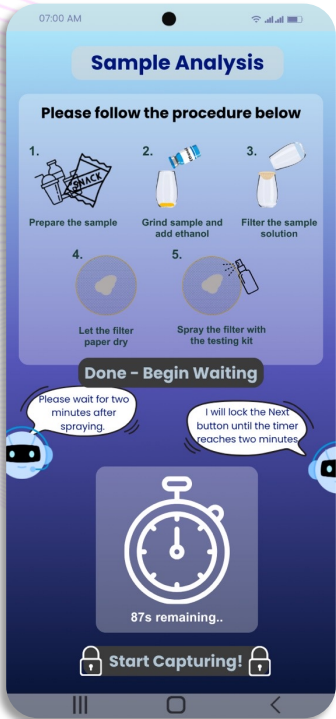
- SPOT-MP.AI WebApp
- Companion Mobile Interface
- Cloud-Connected Agentic Workflow
- Data Visualization & Awareness Map
- API Integration



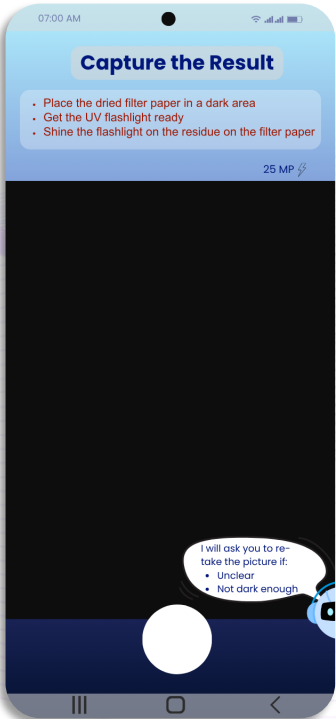
User dashboard displays profile, news, reports, and microplastic awareness maps



Sample entry screen collects key sample details to feed information to Agentic AI



User guidance for microplastic analysis using time-based observation



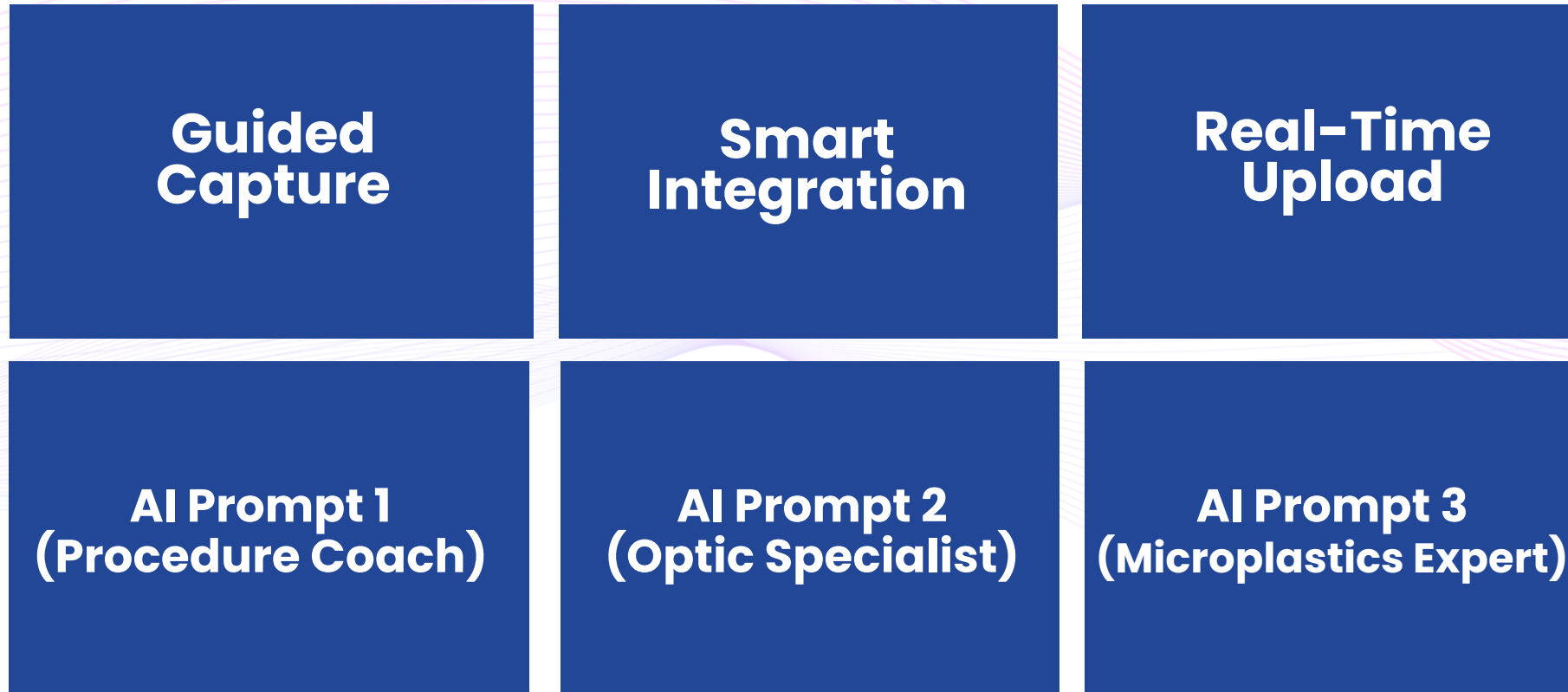
Agentic AI helps to validate photo requirement, clarity, and quality



Real-time assessment summary and seamless data synchronization

# AgenticAI: Your Smart Co-Pilot for Microplastic Detection

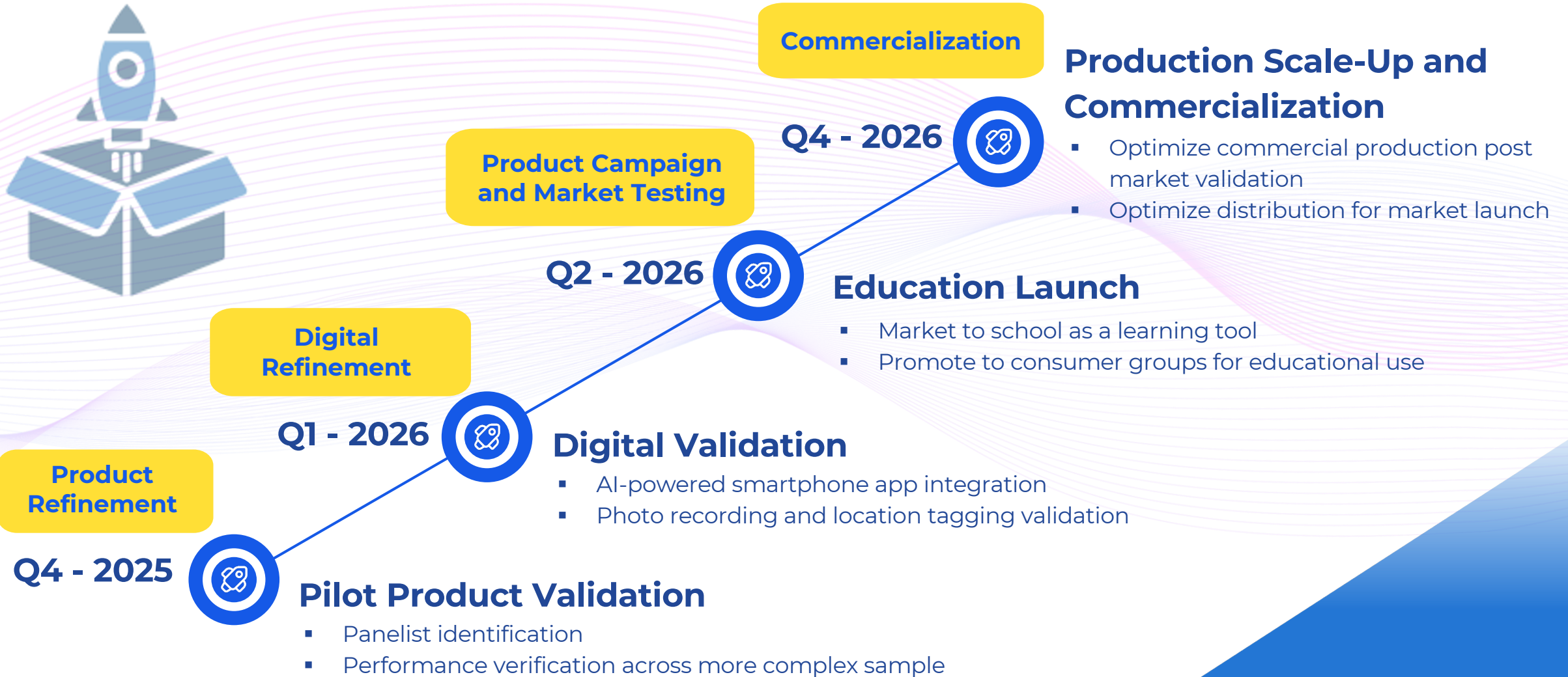
From image capture to expert evaluation, an AI system that guides, learns, and improves with every scan.



AgenticAI transforms every user into a guided scientist, making microplastic detection smart, scalable, and human-friendly.

# SPOT-MP.AI Roadmap

A clear road-map with measurable milestones



# Meet the Team Behind the Innovation

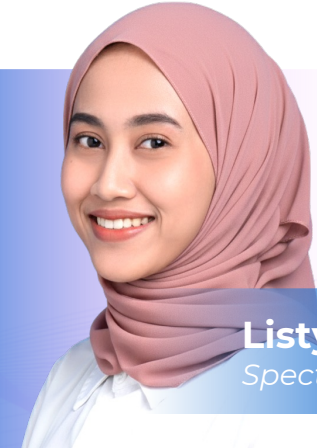
Chemistry. Engineering. AI. One mission: make microplastics visible.



**Nur Allif F.**  
*Vision and Strategy Lead*



**Esra Kotob A.**  
*Sensor & Detection Specialist*



**Listya Eka A.**  
*Spectroscopy & Data Analyst*



**Izzul Qudsi**  
*AI Imaging Engineer*



**Sharhan H. H.**  
*Full-Stack Developer*



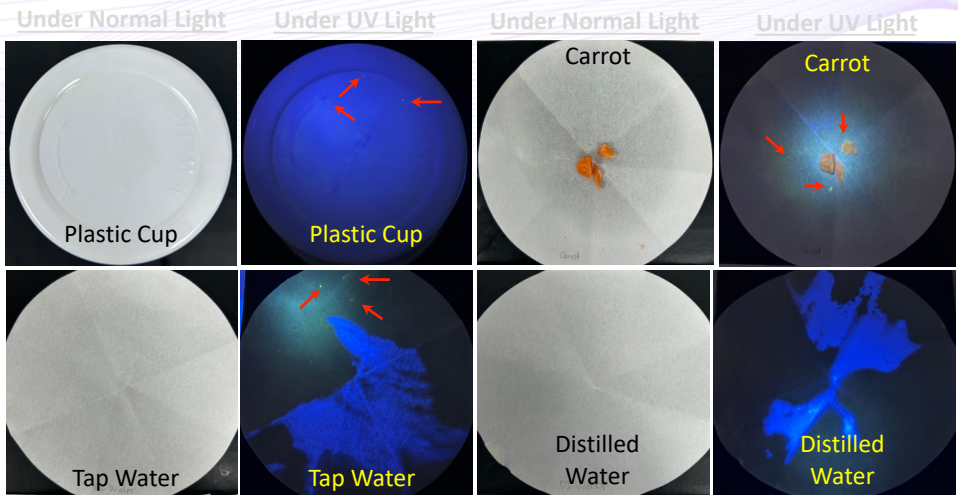
**Anisa Helena I. P.**  
*Design & Outreach Lead*

We are committed to transform invisible pollution into visible awareness.

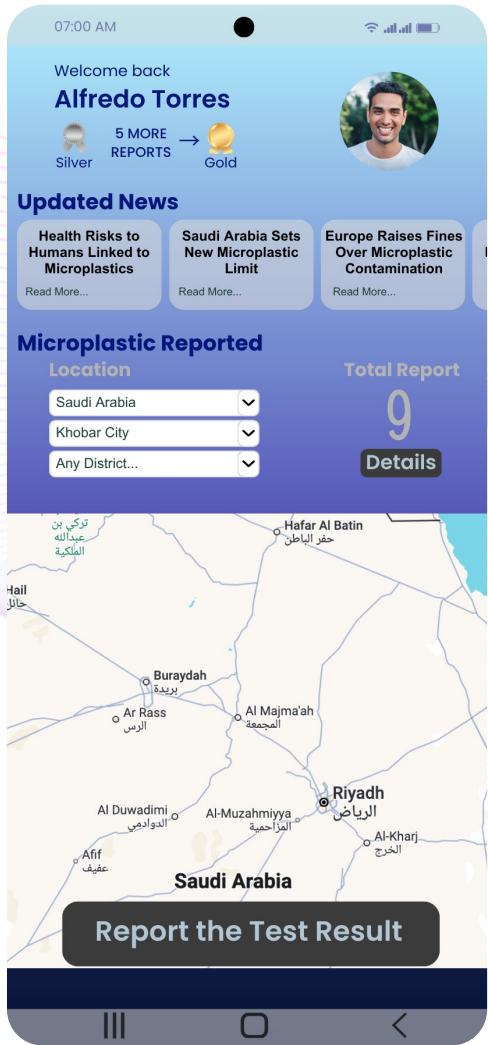
# Product Prototype



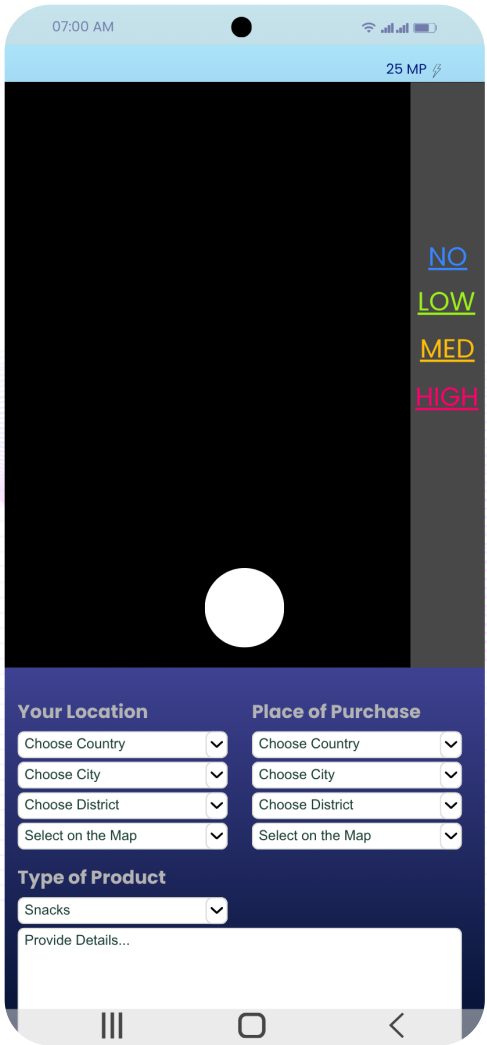
**Non-toxic, bio-based UV microplastic detection kit**



**Test result on the sample**



**Mobile app dashboard**



**Mobile app feature – AI powered microplastic classification and reporting**

# References

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# Thank You.

From a spark of curiosity to global awareness  
*From this Hackathon to the World.*



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Intelligent Microplastic Awareness



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